

Tarea 3.2

$$3) f(x) = (3x^2 - 5x)e^x$$

$$f'(x) = (3x^2 - 5x)e^x + e^x(6x - 5)$$

$$f'(x) = 3x^2e^x - 5xe^x + 6xe^x - 5e^x = e^x(3x^2 + x - 5)$$

$$7) g(x) = \frac{3x - 7}{2x + 7}$$

$$g'(x) = \frac{(2x+7) \cdot 3 - (3x-7) \cdot 2}{(2x+7)^2} = \frac{6x+3-6x+2}{(2x+7)^2} = \frac{5}{(2x+7)^2}$$

$$11) F(y) = \left(\frac{1}{y^2} - \frac{3}{y^4}\right)(y + 5y^3)$$

$$F'(y) = \left(\frac{y^2 - 3}{y^4} \cdot y(1 + 5y^2)\right) = \frac{(y^2 - 3)(1 + 5y^2)}{y^3}$$

$$F'(y) = \frac{y^2 + 5y^4 - 3 - 15y^2}{y^3} = \frac{-14y^2 + 5y^4 - 3}{y^3} = \frac{-14y^2}{y^3} + \frac{5y^4}{y^3} - \frac{3}{y^3} =$$

$$F'(y) = \frac{-14}{y} + 5y - \frac{3}{y^3} = -\frac{14}{y^2} + 5 - \left(-3 \cdot \frac{3y^2}{(y^3)^2}\right)$$

$$F'(y) = -\frac{14}{y^2} + 5 + \frac{9}{y^4}$$

$$23) f(x) = \frac{x^2 e^x}{x^2 + e^x}$$

$$f'(x) = \frac{(x^2 + e^x)(x^2 + e^x) - x^2 e^x (x^2 + e^x)}{(x^2 + e^x)^2} = \frac{(2xe^x + x^2 e^x)(x^2 + e^x) - x^2 e^x (2x e^x)}{(x^2 + e^x)^2}$$

$$f'(x) = \frac{2xe^{2x} + x^4 e^x}{(x^2 + e^x)^2} = \frac{xe^x(2e^x + x^3)}{(x^2 + e^x)^2}$$

$$\frac{1}{1+e^x}$$

$$f'(x) = \frac{(1+e^x)2x - (x^2)e^x}{(1+e^x)^2} = \frac{2x + 2xe^x - x^2e^x}{(1+e^x)^2} = \frac{x(2 + 2e^x - xe^x)}{(1+e^x)^2}$$

$$f''(x) = \frac{x(2 - 2e^x - xe^x)}{(1+e^x)^2} = \frac{2x - 2xe^x - x^2e^x}{(1+e^x)^2} =$$

$$f'''(x) = \frac{(2x - 2xe^x - x^2e^x)(1+e^x)^2 - (2x - 2xe^x - x^2e^x)(1+e^x)^2}{((1+e^x)^2)^2} =$$

$$f'''(x) = \frac{(2 - 2e^x - 2xe^x - x^2e^x)(1+e^x)^2 - (2x - 2xe^x - x^2e^x)2(1+e^x)e^x}{((1+e^x)^2)^2}$$

$$f'''(x) = \frac{2 - 2e^{2x} - 8xe^x - x^2e^x + x^2e^{2x}}{(1+e^x)^3}$$

$$37) y = \frac{x^2 - 1}{x^2 + x + 1} \quad (1, 0)$$

$$y' = \frac{(x^2 + x + 1)(2x) - (x^2 - 1)(2x + 1)}{(x^2 + x + 1)^2} = \frac{2x^3 + 2x^2 + 2x - (2x^3 + x^2 - 2x - 1)}{(x^2 + x + 1)^2}$$

$$y' = \frac{x^2 + 4x + 1}{(x^2 + x + 1)^2}$$

$$y'|_1 = \frac{(1)^2 + 4(1) + 1}{(1^2 + 1 + 1)^2} = \frac{6}{9} = \frac{2}{3}$$

$$\frac{y}{y} = \frac{0}{\frac{2}{3}x - \frac{2}{3}} = \frac{\frac{2}{3}(x-1)}{\frac{2}{3}x - \frac{2}{3}}$$

$$33) y = 2xe^x$$

$$(0, 0)$$

$$y' = (2x)e^x + 2x(e^x) = 2e^x + 2xe^x$$

$$y'|_0 = 2e^0 + 2(0)e^0 = 2$$

$$\tan = y - 0 = 2(x - 0)$$

$$y = 2x$$

$$\text{normal} = y - 0 = -\frac{1}{2}(x - 0)$$

$$y = -\frac{1}{2}x$$

Scribe